



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic business decisions

Dataset at a Glance

3,900

Total Purchases

Transactions analyzed
across all categories

18

Data Points

Features tracking
demographics and
behavior

25

Product Types

Items across 4 major
categories

50

Locations

Geographic coverage
nationwide



Data Preparation Journey

01

Data Loading & Exploration

Imported dataset, analyzed structure and summary statistics

02

Cleaning & Standardization

Handled missing values, renamed columns to snake case

03

Feature Engineering

Created age groups and purchase frequency metrics

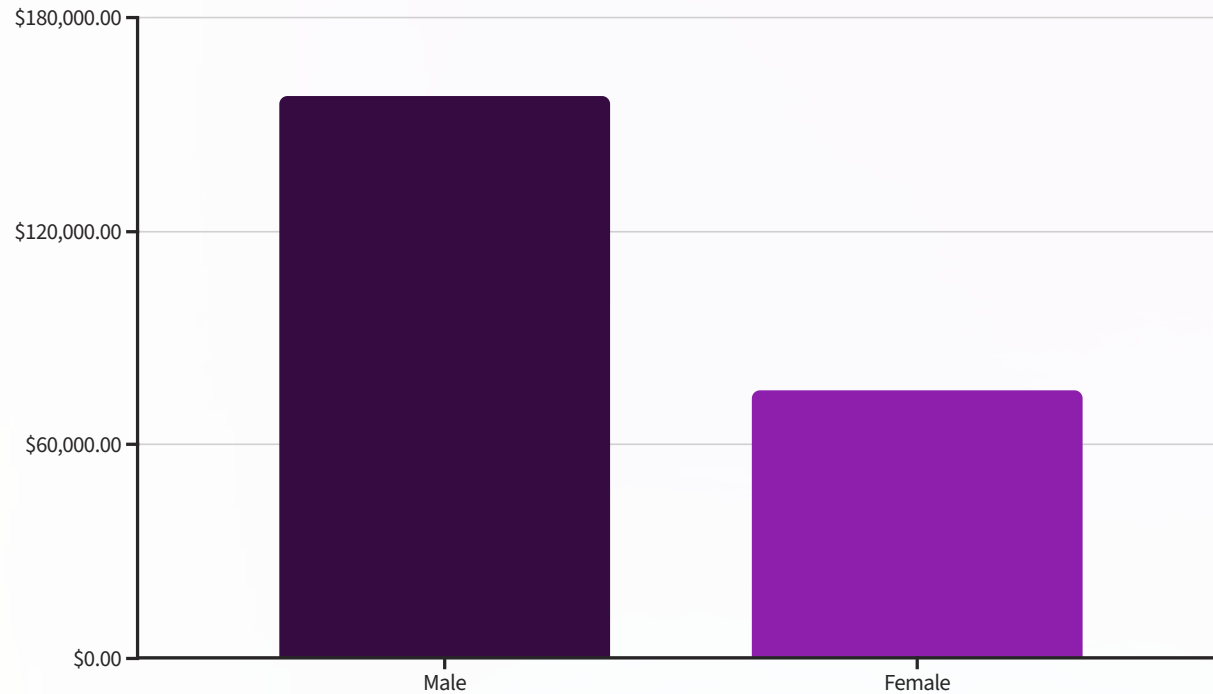
04

Database Integration

Connected to PostgreSQL for advanced SQL analysis



Revenue Insights



Gender Revenue Gap

Male customers generate 2.1x more revenue than female customers

Opportunity for targeted marketing to balance customer base



Smart Shoppers

839 Customers

Used discounts but spent above average (\$60)

High-Value Buyers

Discount-sensitive yet willing to spend more

Strategic Opportunity

Target with premium product promotions

Top-Rated Products

1

Gloves

3.86 average rating

2

Sandals

3.84 average rating

3

Boots

3.82 average rating

4

Hat

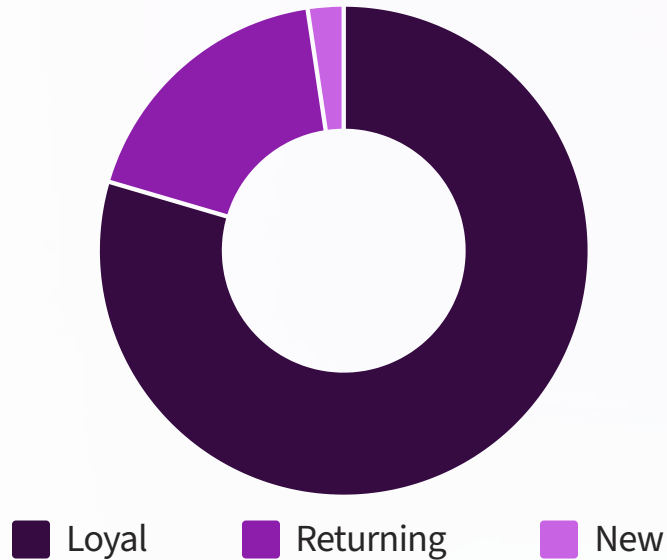
3.80 average rating

5

Skirt

3.78 average rating

Customer Loyalty Breakdown



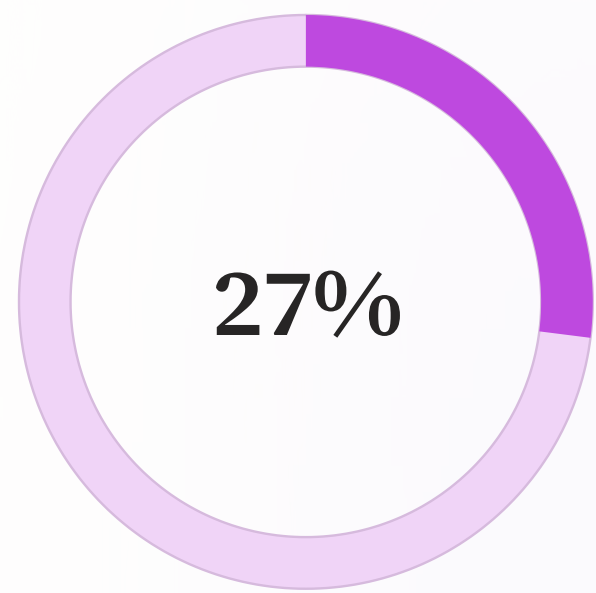
Strong Loyalty Base

80% of customers are loyal buyers

Focus on converting returning customers to loyal status

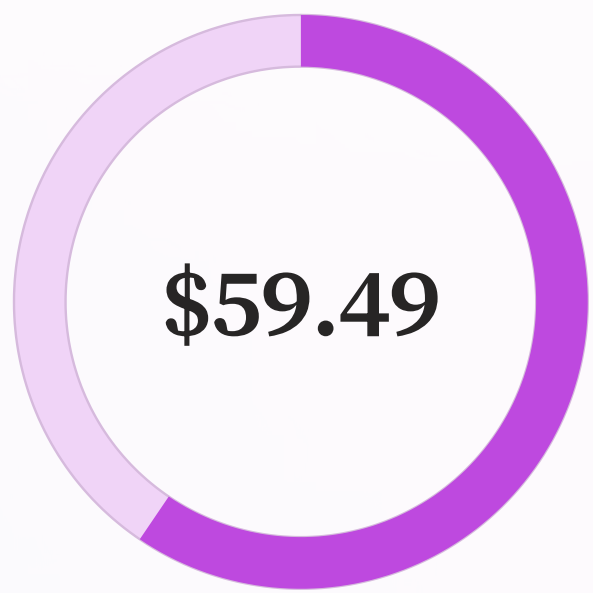
958 repeat buyers (>5 purchases) show subscription potential

Subscription Analysis



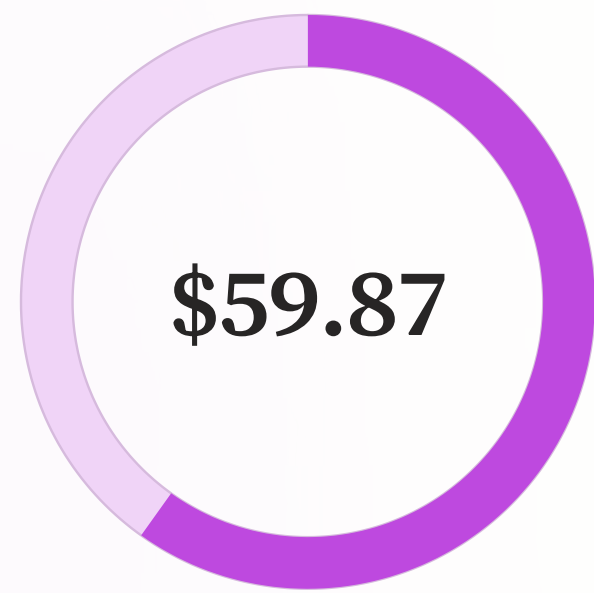
Subscription Rate

1,053 of 3,900 customers subscribed



Subscriber Spend

Average purchase amount



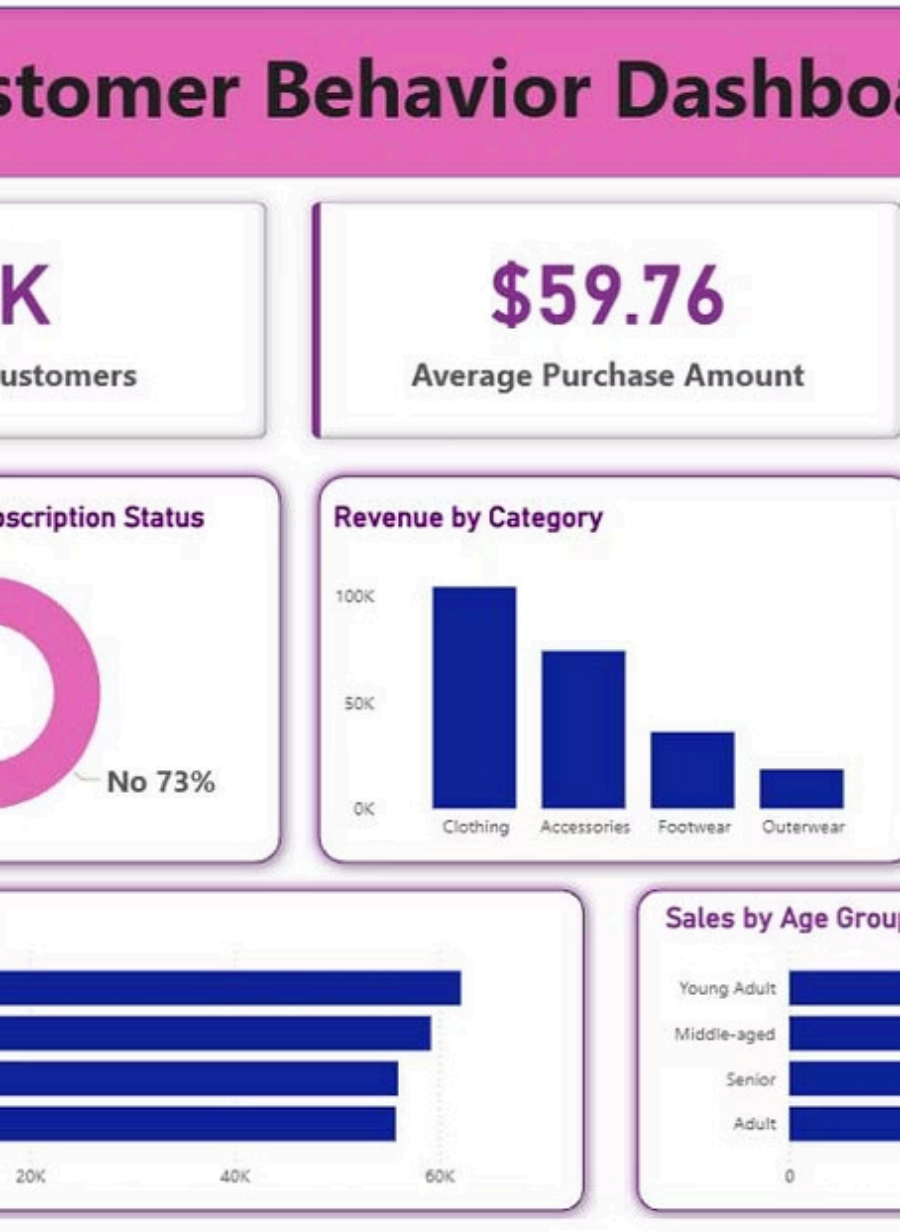
Non-Subscriber Spend

Minimal difference in spending

Shipping Preferences

- Express: \$60.48 average
- Standard: \$58.46 average
- Express users spend 3.5% more

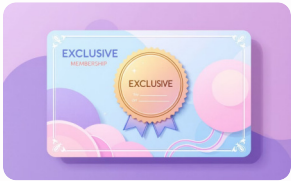




Interactive Dashboard

Power BI dashboard visualizing key metrics: revenue trends, customer segments, product performance, and behavioral patterns

Strategic Recommendations



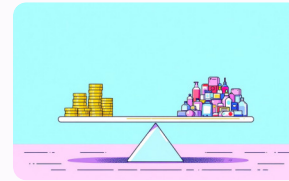
Boost Subscriptions

Promote exclusive benefits to convert repeat buyers



Loyalty Programs

Reward repeat buyers to increase retention



Optimize Discounts

Balance sales growth with margin control



Targeted Marketing

Focus on high-revenue age groups and express shipping users